

Baitong (Barbara) Mu

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EDUCATION

Wake Forest University

MS in Psychology

08/2025 - 05/2027 (Expected)

North Carolina, US

University of Edinburgh

BSc. (Hons) Psychology - 4 Years | GPA: 3.61/4.00

09/2021 - 05/2025

Edinburgh, Scotland, UK

GRE: 333(V163-91%; Q170-92%) + AW4.5-83%

RESEARCH INTERESTS

Cognitive and Neural Mechanisms of Visual Attention, Working Memory, Event Memory; Eye-tracking, Neuroimaging, Computational Modeling

RESEARCH OUTPUTS

- **Mu, B.** & Stilwell, B. T. (In Preparation). *Suppression without overt selection: Learned distractor suppression does not depend on initial oculomotor capture.*
- **Mu, B.** & Stilwell, B. T. (2026, November). Triggered Distractor Suppression: Learned Suppression Without Initial Overt Capture. Poster presentation accepted at the 67th Annual Meeting of the Psychonomic Society and the 34th Object Perception, Attention, and Memory (OPAM) Conference, San Diego, CA, US.
- Sun, H., **Mu, B.**, & Sali, A. W. (2026, November). Attentional Flexibility as an Internal Context: A Temporally Graded Context-Dependent Priming Effect. Poster presentation accepted at the 34th Object Perception, Attention, and Memory (OPAM) Conference, San Diego, California, US.
- **Mu, B.** & Stilwell, B. T. (2026, May). Generalization of Learned Distractor Suppression. Poster presentation at the Vision Sciences Society Annual Meeting, Tampa, FL, US.
- **Mu, B.** & Stilwell, B. T. (2026, March). Skill-based Learning in Distractor Suppression Across Visual Context. Poster presentation at the 2026 North Carolina Cognition Conference (NCCC), Durham, NC, US.
- Sun, H., **Mu, B.**, & Sali, A. W. (2026, March). Attentional States as Psychological Context for Long-term Memory: An Examination of Encoding Specificity. Poster presentation at the 2026 North Carolina Cognition Conference (NCCC), Durham, NC, US.
- **Mu, B.**, Ansari, S., Xi, Y., & Lawrence, A. D. (2025, March). Cultural Variation in Episodic Memory and Event Cognition. Oral presentation at the British Psychological Society Scottish Undergraduate Conference, Dundee, UK.
- Sanaae, S.[†], **Mu, B.**[†], Tang, C. K.-M., Caslick-Waller, Z. R. G., Li, W., Gao, T., & Rodriguez Cruces, R. (2025). Parallel GNN-LSTM model predicting working memory involvement during language and emotion processing. Neuromatch. ([†]Co-first authors; <https://doi.org/10.5281/zenodo.15126506>)

RESEARCH EXPERIENCE

Graduate Researcher

Advisors: Dr. Brad T. Stilwell and Dr. Anthony W. Sali

09/2025 – present

Wake Forest University

Attentional Control Across Learning, Perception, and Memory

- Research focus: Investigated how experience shapes attentional control, including how attentional priorities are acquired and generalized across contexts, how task-relevant information is maintained or transformed after guiding selection, and how prior attentional states influence subsequent memory and behavior.
- Methods: Designed behavioral and eye-tracking experiments using visual search, gaze-contingent, RSVP, attentional shift/hold, and continuous-report paradigms; analyzed behavioral, fixation, and continuous-report data using mixed-effects models, Hidden Markov models, and hierarchical Bayesian mixture modeling with parameter-recovery simulations.
- Current work: Examining how negative attentional templates are maintained or transformed in visual working memory after guiding visual search using eye tracking, continuous report, and Bayesian mixture modeling.
- Skills: Experimental design, Eye-tracking methods, Bayesian mixture modeling, MATLAB/Psychtoolbox, R.

Research Intern

Advisor: Prof. Roberto Cabeza and Dr. Maxi Becker

06/2026 – present

Duke University

Project: Encoding Gaze Dynamics and Subsequent Memory for Perceptually Ambiguous Images

- Research question: How does active sampling of diagnostic evidence support the transition from perceptual uncertainty to recognition of two-tone (Mooney) images, and how does this process relate to subsequent memory?
- Methods: Developed a web-based ROI annotation interface and derived gaze-independent diagnostic regions from 7 human raters with inter-rater reliability analyses; Analyzed pre-solution fixation dynamics, gaze similarity across time windows and response conditions (*eyesim*), image-level saliency (SaliencyToolbox) and contour complexity (ImagingToolbox), as predictors of recognition and memory success.
- Interim Results: Early diagnostic sampling predicted recognition, whereas late gaze convergence tracked decision commitment rather than accuracy; early diagnostic sampling showed no direct effect on 5-day delayed memory, with its association with memory mediated by successful recognition.
- Skills: Gaze pattern analysis, time-resolved gaze similarity analysis, computational image analysis, Python, R, MATLAB.

Research Intern

05/2026 – present

Advisor: Dr. Maureen Ritchey and Dr. Valentina Krenz

University of California, Los Angeles (Remote)

Project: Posterior Medial Contributions to Event Transitions and Subsequent Memory

- Research question: Does posterior medial (PM) network engagement at event transitions during encoding predict the quality of subsequent memory for multi-featural events?
- Methods: Reviewed MRIQC and fMRIPrep outputs to identify artifacts and assess head motion for quality assessment and control; Established motion-based exclusion criteria for the dataset, including diagnostics for task-correlated motion in overt-recall runs; Authored a step-by-step QC protocol documenting the full pipeline for lab-wide use
- Current work: Implementing univariate and multivariate (regularized regression) subsequent-memory analyses testing whether hippocampal and PM offset activity dissociate the binding of event features from their mnemonic specificity.
- Skills: fMRI quality control, multivariate analysis, reproducible protocol development, MRIQC, fMRIPrep, Jupyter Notebook.

Undergraduate Dissertation Project

Advisor: Prof. Andrew D. Lawrence

09/2024 – 04/2025

Final-Year Dissertation Project: Cultural Variation in Event Segmentation and Episodic Memory

- Research question: How does cultural variation in thinking style and self-construal influence event perception and episodic memory during naturalistic movie viewing?
- Methods: Built an event-segmentation, free-recall, and cultural-measure protocol (Analysis-Holism Scale; Auckland Individualism and Collectivism Scale) in Qualtrics with JavaScript around an 8-minute movie *Bang! You're Dead*; conducted path mediation analysis to explore the serial mediation relationship
- Results: No significant cultural-group differences in event segmentation and memory; holistic thinking styles were associated with reduced recall details was found in exploratory analyses
- Skills: Experiment design, Naturalistic event-segmentation paradigm, Cross-cultural research, R, Path mediation analysis

NeuroMatch Academy & Impact Scholar Program

Advisor: Dr. Raul Rodriguez Cruces

07/2024 – 05/2025

Project: Working-Memory Network Generalization Across Cognitive Domains

- Research question: How does WM neural network involvement generalize across cognitive domains, including language (story comprehension vs. math language) and emotion (neutral vs. fear)
- Methods: Built a Multilayer Perceptron (MLP) classifier on HCP fMRI data to classify WM load (0-back vs. 2-back); Assessed cross-task generalization by inputting BOLD signals from language and emotion subtasks into the whole-brain and frontoparietal networks; extended the pipeline into a parallel GNN-LSTM model to characterize the frontoparietal and default mode networks dynamics during high-level cognitive processing
- Results: MLP classifier achieved 81% accuracy; WM demand classification generalized to language tasks, with math stimuli predicted as high demand and story-based stimuli as low demand; classification was less confident in emotion subtasks, potentially due to the rapid amygdala-mediated processing reducing reliance on the executive WM system
- Skills: Python, MLP, LSTM, GNN, fMRI analysis (GLM), and HCP dataset

Ad Hoc JOURNAL REVIEWING

Journal of Experimental Psychology: Human, Perception, & Performance

SCHOLARSHIP & AWARDS

Edinburgh Award (Student Development)

04/2025

Wake Forest University Graduate Fellowship (\$61,145 per year)

08/2025 – 05/2027

Professional Development Award, Object Perception, Attention, and Memory (OPAM) Conference (\$500)

11/2026